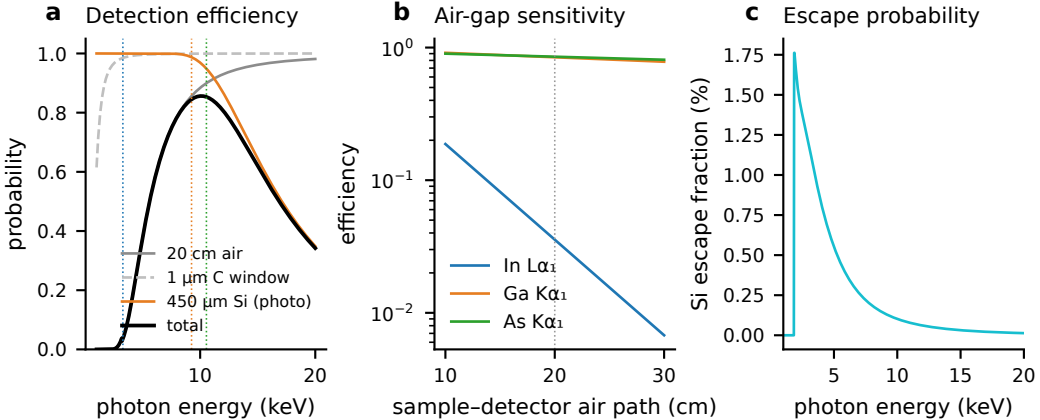
a
Detection efficiency
probability
1.0
0.8
0.6
0.4
0.2
0.0
10
20
photon energy (keV)
20 cm air
1 µm C window
450 µm Si (photo)
total
b
Air-gap sensitivity
efficiency
10^{0}
10^{-1}
10^{-2}
10
20
30
sample–detector air path (cm)
In Lα₁
Ga Kα₁
As Kα₁
c
Escape probability
Si escape fraction (%)
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
5
10
15
20
photon energy (keV)